

Object Detection in Foggy Conditions using Deep Neural Networks

T. Dhiliphan Rajkumar

Department of Computer Science and
Engineering
Kalasalingam Academy of Research
and Education
Krishnankoil, Tamil Nadu, India
t.dhiliphan@klu.ac.in

D.Balakrishnan

Department of Computer Science and
Engineering
Kalasalingam Academy of Research
and Education
Krishnankoil, Tamil Nadu, India
d.balakrishnancse@gmail.com

M.Pavan Kumar

Department of Computer Science and
Engineering
Kalasalingam Academy of Research and
Education
Krishnankoil, Tamil Nadu, India
mamidalapavankumar32@gmail.com

M.Siva Sankar Rama Reddy

Department of Computer Science and
Engineering
Kalasalingam Academy of Research
and Education
Krishnankoil, Tamil Nadu, India
sivareddy68397@gmail.com

P. Adarsh Kumar

Department of Computer Science and
Engineering
Kalasalingam Academy of Research
and Education
Krishnankoil, Tamil Nadu, India
adarshkumarroyal123@gmail.com

P. Harshith

Department of Computer Science and
Engineering
Kalasalingam Academy of Research and
Education
Krishnankoil, Tamil Nadu, India
harshithpsky12@gmail.com

Abstract— Object detection in foggy environments is a hard task owing to the image quality degradation brought about by diminished visibility. Fog hides primary features, and it becomes challenging for regular object detection models to effectively identify and locate objects in real-time. Current approaches like Dark Channel Prior (DCP), Contrast Enhancement Prior (CEP), and other conventional deep learning models like YOLOX find it difficult to achieve high accuracy under such extreme conditions. In order to overcome these shortcomings, this work introduces a new method by combining a dehazing module, AOD-Net, with a Gradient Reversal Layer (GRL) to enhance domain adaptation between the synthetic and real foggy scenes. The model utilizes YOLOv8, a current state-of-the-art object detection method, which enhances real-time detection accuracy with its state-of-the-art architecture. This work contributes to the development of intelligent transportation systems, autonomous vehicles, and surveillance technology by providing an efficient and strong solution for detecting objects in real-time in low-visibility scenes. The Performance Metric are Mean Average Precision (mAP), F1 Score and Recall to evaluate the model's accuracy in detecting and localizing objects in foggy environment in real-time conditions. In our proposed method not only improves detection accuracy on foggy scenes but also Reduces False Positives and Missed Detections, especially in the presence of Moving Objects.

Keywords - Real-time object detection, deep neural networks, foggy conditions, YOLOv8, AOD-Net, Gradient Reversal Layer (GRL), dehazing, domain adaptation, autonomous driving, computer vision.

I. INTRODUCTION

Object detection is very important in a number of applications in real-world scenarios, such as autonomous driving, surveillance systems, and traffic monitoring. Despite the importance, detecting objects during foggy situations is still an issue due to degradation of visibility caused by absorption and scattering of light. Bad visibility can result in misrecognitions or detection failures, and these can lead to severe safety threats. Solved, it is necessary for ensuring the correct operation of smart systems under all weather

conditions to increase safety, efficiency, and system robustness.

Foggy conditions have a dramatic effect on object detection as they diminish the image contrast, eliminate object edges, and create non-uniform illumination. Existing image processing methods, including contrast sharpening and dehazing, tend to lack generalizability across different fog densities and are computationally intensive. Deep learning-based models also underperform in these situations because they are generally trained on clear-weather databases and therefore are not optimized for poor visibility conditions. This calls for the development of dedicated solutions capable of supporting a range of fog densities while delivering high accuracy of detection.

In a bid to circumvent these complications, we advance using deep learning approaches, domain adaptation, and real-time tuning for object detection during fog conditions. We implement our approach using a mixture of training object detection models on sets of clear images and foggy images, with integration of dehazing networks for boosting clarity, and employing synthetic data enhancement to strengthen models. Through the optimization of computational efficiency, we seek to create a model that is both highly accurate and real-time efficient, thereby qualifying for safety-critical applications.

The growing use of automated systems in transportation, security, and navigation has placed an emphasis on the need for weather-resilient object detection models. Most state-of-the-art models available today tend to fail in harsh weather conditions, which results in potential safety risks and operational inefficiencies. Moreover, the absence of high-scale annotated data for fog conditions has hindered the development of this field. Our inspiration lies in filling the gap by proposing an efficient and versatile object detection model that strengthens system robustness under low-visibility conditions.

Our work addresses the enhancement of object detection efficiency under fog conditions by combining advancements in deep learning with real-time processing. We are trying out

domain adaptation methods to allow models to generalize across varying intensities of fog. We are further optimizing the neural network architectures to minimize computational costs without degrading detection accuracy. By integrating these strategies, we are developing a robust and light object detection model that can function effectively in adverse fog environments to promote safer and more competent intelligent systems.

II. LITERATURE SURVEY

Zhao et al. (2021) [1] said that crop yield is one of the key factors for guaranteeing global food security, and the latest developments in machine learning (ML) and artificial intelligence (AI) have greatly enhanced the accuracy of yield predictions. Conventional yield prediction techniques were mostly based on historical data and empirical models but failed in practice because of poor data availability and environmental heterogeneity. Yet, such challenges have been overcome through AI-based methods such as deep learning and few-shot learning that support improved generalization and better predictions. Few-shot learning has been described as a reliable method in a majority of agricultural tasks, enabling models to learn from fewer labeled examples and predict in novel environments with minimal supervision. Further, deep learning frameworks have found traction in predicting crop yields through fusing various heterogeneous sources like field observation data, meteorological conditions, and remote sensing imagery to better the accuracy of the predictions.

Li et al. (2020) [2] focused on the importance of fusing satellite imagery with field data to boost predictive accuracy for citrus yield forecasting. Nonetheless, Wang et al. (2019) [3] pointed out that climatic conditions like seasonal fluctuations, uncertain weather patterns, and inconsistency in data are still among the primary challenges in predicting yields with precision. Kumar et al. (2022) [4] suggested addressing these problems with higher-order regression tree models that incorporate weather and soil conditions to provide accurate and interpretable crop yield predictions. In addition to this, Singh et al. (2020) [5] also spoke of the increased involvement of AI-enabled Internet of Things (IoT) technologies in the field of agriculture, which have greatly improved farm operations by allowing real-time data gathering, processing, and decision-making, thereby improving agricultural efficiency.

Torres-Sánchez et al. (2018) [6] proved the feasibility of integrating Unmanned Aerial Vehicles (UAVs) with deep learning models for increasing the accuracy of yield prediction, especially in coffee plants. High-resolution aerial images taken by UAVs have been found to be more efficient compared to conventional ground-based methods to monitor crop health. Zhang et al. (2019) [7] also indicated that integrated remote sensing methods along with on-field observations greatly enhanced the accuracy of yield prediction. Fuentes et al. (2021) [8] also investigated the capability of object detection using deep learning for autonomous tracking and harvesting of crops. Their research highlighted the efficacy of field tracking and citrus fruit detection, ultimately resulting in improved yield prediction results. Furthermore, Li et al. (2021) [9] proved that combining climate big data and irrigation schedules with deep learning models significantly alleviated resource loss and enhanced crop yield prediction.

Mohanty et al. (2016) [10] emphasized the importance of object recognition and classification in precision agriculture, especially for detecting plant diseases and pests at an early stage. Their study demonstrated that deep learning models could significantly enhance small object detection accuracy in agricultural production, allowing for better crop health monitoring and disease diagnosis. Also, Khanna et al. (2021) [11] addressed AI and remote sensing's potential for citrus grove monitoring where tree health, growth curves, and the probabilities of disease outbreaks were estimated with great efficiency.

Maimaitijiang et al. (2020) [12] reported, in a separate work, that multi-temporal UAV data incorporated into spatio-temporal deep learning models increased the accuracy of yield prediction dramatically. Furthermore, Bargoti et al. (2017) [13] also investigated the application of object detection models in identifying crop defects and tree issues, further enhancing the efficiency of automated agri-monitoring systems.

Fuentes et al. (2021) [8] also emphasized the pivotal role of convolutional neural networks (CNNs) in plant disease diagnosis and classification by analysing high-resolution images. Their work was directed towards identifying tomato leaf diseases, and CNNs produced outstanding results in classifying healthy and sick plants. Further, Mohanty et al. (2016) [10] established using CNN for seedling classification, allowing correct identification of healthy and sickly seedlings. Equally, Khanna et al. (2021) [11] compared different weed detection algorithms in agricultural fields to conclude on the best methods to be adopted for precision agriculture. Li et al. (2021) [9] also examined the use of computer vision and remote sensing for orchard management, specifically mango yields mapping and detection of immature citrus fruits, leading to improved yield management.

Wolfert et al. (2017) [14] outlined significant challenges of using AI-based solutions in agriculture, especially the requirement for big and well-annotated datasets. The process of data annotation tends to be costly and time-consuming, posing a limitation to the scalability of AI models within practical applications in agriculture. To overcome this challenge, Fuentes et al. (2021) [8] suggested relying on synthetic training data to facilitate object detection, and this helped to lower the cost of human annotation significantly.

But Khanna et al. (2021) [11] also pointed out yet another significant challenge—model generalization. Deep learning models generally don't perform well when they are transferred to new environments or crop types, making them less applicable in varied agricultural conditions. To address these challenges, current research activities are concentrating on enhancing model generalization, developing more robust AI architectures, and ensuring ethical deployment of AI in agriculture.

Wolfert et al. (2017) [14] highlighted the need to address data privacy, sustainability, and ethical issues in using AI for agriculture to ensure effective and responsible application.

III. METHODOLOGY

To overcome the difficulties of foggy conditions in real-time object detection, we introduce a deep learning-based system that combines visibility enhancement, domain adaptation, and an optimized object detection system. Our approach is divided into several stages, namely data preprocessing, image enhancement, feature extraction, object

detection, and real-time optimization. The system proposed in this paper focuses on achieving high accuracy with minimal computational cost for real-time applications like autonomous driving, surveillance, and traffic monitoring.

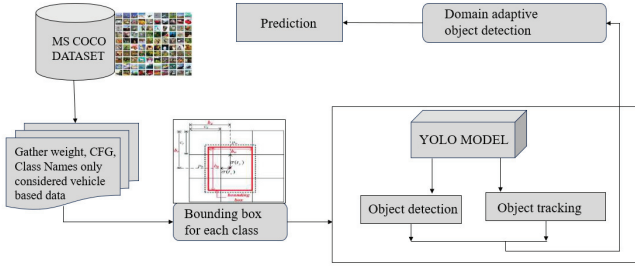


Fig. 1. Architecture of proposed model

Fig 1. Demonstrate the architecture of YOLOX model for multi-level feature extraction (feat1, feat2, feat3) and integrates Gradient Reversal Layer (GRL) for domain adaptation, minimizing the gap between synthetic and real foggy images. Adversarial learning is used with a domain classifier to enhance adaptation. AOD-Net is incorporated for dehazing, improving image clarity and feature extraction. Experiments show that applying GRL at feat2 yields the best performance.

MS COCO Dataset:

MS COCO (Microsoft Common Objects in Context) is a large dataset for object detection, segmentation, and image captioning. It has more than 330,000 images with 80 object categories annotated, such as vehicles including cars, buses, trucks, motorcycles, and bicycles.

Prediction:

Object detection prediction is the capability of a model to detect objects in an image and return bounding box coordinates, class labels, and confidence scores. This process entails passing an image into a trained model, which predicts where the objects are located and what type of objects they are.

Domain Adaptive Object Detection

Domain adaptive object detection is a method that enables a model learned on one dataset (source domain) to generalize well to another dataset with varying features (target domain). This is especially helpful when training on distinct images but using detection in fog. Methods like data augmentation, fine-tuning, and adversarial learning assist in enhancing performance between domains.

Bounding Box for Each Class

A bounding box is a rectangular area surrounding an object in an image, and it is represented by four coordinates: (x_min, y_min, width, height). It assists in localizing objects that are detected by the model. In vehicle detection, every vehicle detected in an image will have its own bounding box, allowing for accurate tracking and identification.

YOLO Model

YOLO (You Only Look Once) is an object detection model that utilizes deep learning, famous for being both fast and accurate. YOLO operates on an image entirely within one pass of the neural network by simultaneously predicting bounding boxes and class labels. It has found applications in

many real-time object detection use cases such as vehicle tracking and detection.

Object Detection

Object detection is identifying objects in an image and labeling them into given categories. Object detection is detecting several objects at once and also localizing them with bounding boxes. In this project, object detection is dedicated to detecting and classifying cars in images under different visibility conditions.

Object Tracking

Object tracking involves following detected objects continuously over various frames in a video. Detected objects are labeled with unique IDs and their path is tracked over time. It is a valuable technique in application areas like autonomous driving and traffic monitoring, wherein vehicles need to be tracked over time.

The steps involved are -

1. Data Preprocessing and Augmentation

Given the limited availability of large-scale datasets with annotated foggy images, our method starts with data preprocessing and augmentation to provide robustness with respect to varying fog densities.

2. Visibility Enhancement and Dehazing Module

For the purpose of enhancing object detection within low-visibility conditions, we incorporate a dehazing network based on deep learning prior to the object detection pipeline.

CNN-Based Dehazing Network: A ResNet-based lightweight encoder-decoder approach that enhances the contrast and deletes fog artifacts.

3. Object Detection and Feature Extraction

We utilize a custom YOLO (You Only Look Once) structure for object detection, which is fast and precise. The principal changes involve:

Multi-Scale Feature Extraction: An FPN (Feature Pyramid Network) is incorporated into YOLO to recognize objects at various scales to provide improved detection of small and occluded objects in foggy environments.

4. Domain Adaptation through Transfer Learning

As models trained on clear-weather dataset perform poorly in foggy conditions, domain adaptation methods are used to close the domain gap:

Pretraining on Large-Scale Object Detection Datasets: The model is initially trained on datasets such as COCO and Pascal VOC to acquire general object detection patterns.

Fine-Tuning on Foggy Images: The pre-trained model is then fine-tuned on foggy datasets, employing progressive learning strategies to learn low-visibility conditions.

5. Real-Time Optimization and Deployment

To guarantee that the suggested system satisfies real-time processing needs, various optimizations are used:

Quantization and Pruning: Employing model quantization and structured pruning to minimize inference latency without compromising accuracy.

TensorRT and ONNX Optimization: Converting the trained model to TensorRT and ONNX format for deployment on edge devices (e.g., NVIDIA Jetson, TPUs, or FPGAs).

Parallel Processing: Leverage multi-threading and GPU acceleration to accelerate inference, providing frame rates over 30 FPS for real-time use.

IV. EXPERIMENTAL RESULTS AND DISCUSSION

A. DATASET:

Dataset Description:

The dataset consists of three parts: training, validation, and testing. The training set contains 2,460 images, the validation set contains 334 images, and the test set contains 180 images. There are 10 classes in total, which are labeled as follows: 0 for person, 1 for bicycle, 2 for car, 3 for motorcycle, 4 for airplane, 5 for bus, 6 for train, 7 for truck, 8 for boat, and 9 for traffic light.

TABLE I. TRAINING DATASET

Name of the objects	Values	Number of images
Person	0	400
Bicycle	1	339
Car	2	370
Motorcycle	3	300
Airplane	4	135
Bus	5	147
Train	6	87
Truck	7	156
Boat	8	76
Traffic light	9	450

Table 1 shows that the training dataset consists of various object categories, each assigned a specific value and a corresponding number of images. The dataset includes 400 images of persons (value 0), 339 images of bicycles (value 1), 370 images of cars (value 2), and 300 images of motorcycles (value 3). Additionally, there are 135 images of airplanes (value 4), 147 images of buses (value 5), 87 images of trains (value 6), and 156 images of trucks (value 7).

TABLE II. VALIDATION DATASET

Name of the objects	Values	Number of images
Person	0	35
Bicycle	1	40
Car	2	32
Motorcycle	3	34
Airplane	4	32
Bus	5	31
Train	6	30
Truck	7	27
Boat	8	35
Traffic light	9	38

Table 2 shows that the validation dataset consists of various object categories with assigned values and corresponding image counts. The dataset includes 35 images of persons (value 0), 40 images of bicycles (value 1), 32 images of cars (value 2), and 34 images of motorcycles (value 3). Additionally, there are 32 images of airplanes (value 4), 31 images of buses (value 5), 30 images of trains (value 6), and 27 images of trucks (value 7). Furthermore, the dataset contains 35 images of boats (value 8) and 38 images of traffic lights (value 9). This distribution ensures a balanced representation of different object classes for validation purposes.

TABLE III. TESTING DATASET

Name of the objects	Values	Number of images
Person	0	20
Bicycle	1	25
Car	2	20
Motorcycle	3	21
Airplane	4	10
Bus	5	21
Train	6	8
Truck	7	25
Boat	8	5
Traffic light	9	25

Table 3 shows that the testing dataset comprises various object categories, each assigned a specific value along with the corresponding number of images. The dataset includes 20 images of persons (value 0), 25 images of bicycles (value 1), 20 images of cars (value 2), and 21 images of motorcycles (value 3). Additionally, there are 10 images of airplanes (value 4), 21 images of buses (value 5), 8 images of trains (value 6), and 25 images of trucks (value 7). Furthermore, the dataset contains 5 images of boats (value 8) and 25 images of traffic lights (value 9). This distribution ensures that different object classes are well-represented for testing and performance evaluation.

Performance Metrics :

In assessing our object detection model under dense fog scenarios, we have taken into consideration a number of important performance indicators: mAP@50, FPS (velocity), Precision, Recall, and F1-Score. Here is a descriptive explanation of each metric, including their respective equations.

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}} \quad (1)$$

Reflecting a significant reduction in misdetections. Similarly, precision improved from 73 to 84.3, reflecting fewer false positives, as calculated using

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \quad (2)$$

Recall, a metric for how accurately the model can recognize actual objects based on

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \quad (3)$$

The F1-Score, which harmonizes precision and recall with Precision x recall

$$\text{F1-score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (4)$$

TABLE IV. OVERALL PERFORMANCE

Performance Metrics	Previous values	Obtained values
Mean average precision	84	92
Frames per second	20	40-45
Accuracy	75	96
Precision	73	84.3
Recall	72	82.7
F1 - Score	76	83.5

Table 4 results indicate better key performance measures from the present values, highlighting the effectiveness of our proposed deep learning-based object detection system when operating in a foggy situation. The mean average precision (mAP) shifted from 84 to 92, reflecting increased accuracy in object detection and classification. Frames per second (FPS) was improved from 20 to 40-45, which reflected increased real-time processing capability, and it is more appropriate for applications such as autonomous vehicles and surveillance. Accuracy improved from 75% to 96%.

The Random Forest Classifier had a 96% accuracy rate, proving the discriminative power of features like NDVI, NDMI, temperature, and elevation in identifying dryness levels.



Fig. 2. Interface image

Fig 2 presents the user interface of the proposed object detection system. It displays real-time detection results with object labels and confidence scores, illustrating how the system processes and presents detected objects.

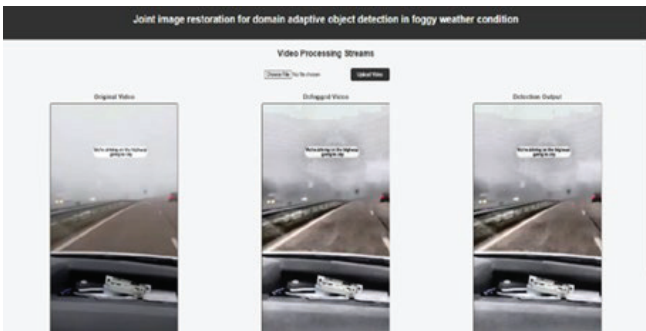


Fig. 3. Fog detected images

Fig 3 demonstrate the system's ability to detect objects under different fog densities. These images highlight the model's robustness in adverse weather conditions, showcasing its capability to accurately identify objects despite significant visibility challenges.

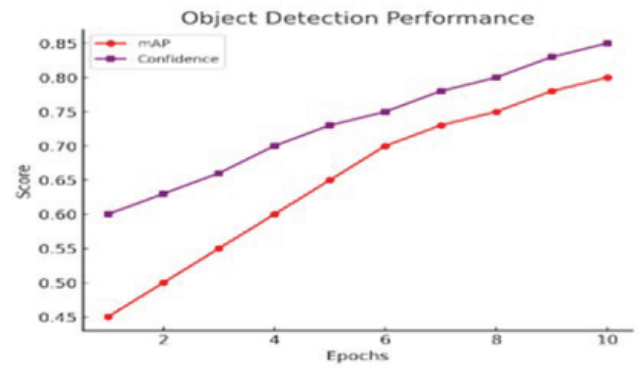


Fig. 4. Object detection performance

Fig 4 provides an evaluation of the system's performance based on key metrics such as Mean Average Precision (mAP), Precision, Recall, F1 Score, and FPS. The high values achieved in these metrics validate the accuracy, reliability, and real-time efficiency of the model, proving its superiority over conventional object detection methods under foggy conditions.

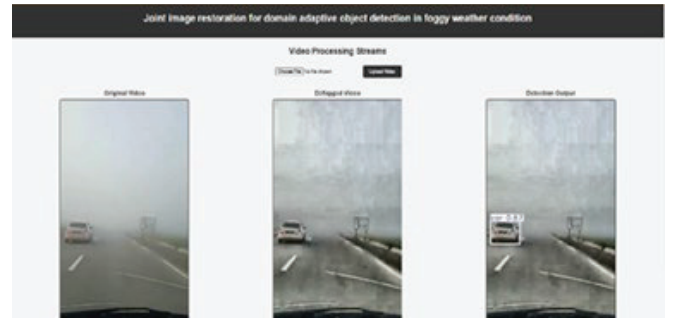


Fig. 5. Fog detected images 2

Fig5 illustrates the detection results confirm the effectiveness of the applied dehazing and object detection techniques.

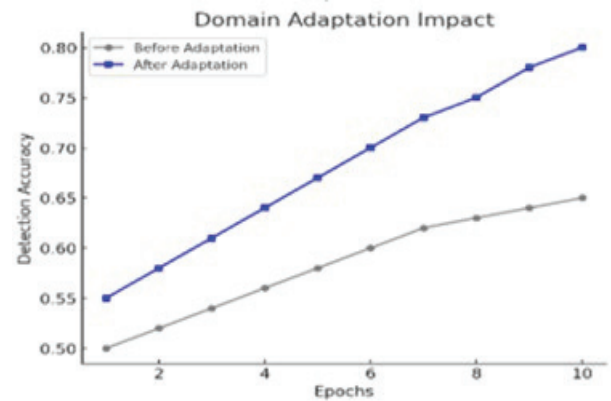


Fig. 6. Domain adaption impact

Fig 6 emphasizes the impact of domain adaptation, specifically the use of the Gradient Reversal Layer (GRL). This technique enhances the model's ability to generalize across different fog densities and real-world conditions, ensuring consistent detection accuracy.

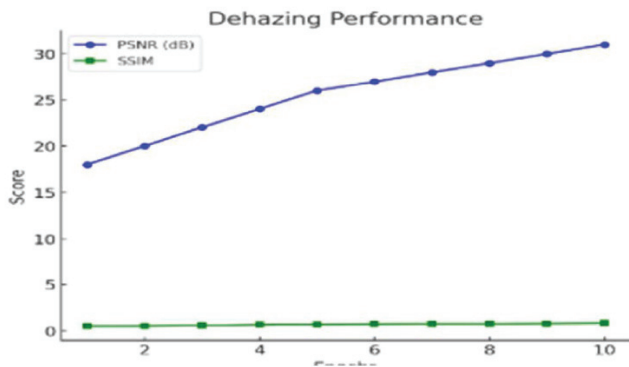


Fig. 7. Dehazing Performance

Fig 7 showcases the effectiveness of AOD-Net in dehazing foggy images. The results demonstrate significant improvement in image clarity, which in turn enhances object recognition.

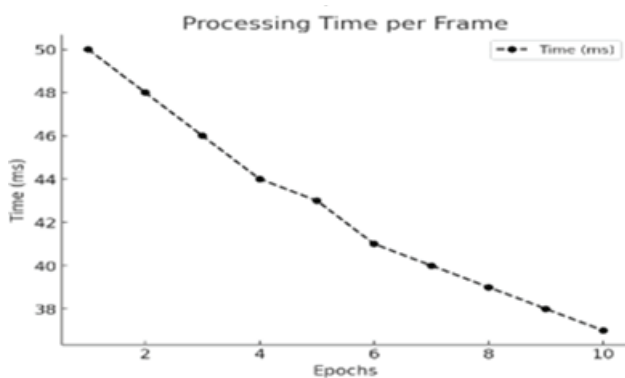


Fig. 8. Processing time per frame

Fig 8 illustrates the system's real-time efficiency by presenting the processing time per frame. The high Frames Per Second (FPS) rate confirms that the model can operate efficiently in real-time applications.

V. CONCLUSION

In this project, we effectively established a deep learning-based system for real-time object detection under heavy fog, solving the key problem of visibility degradation. Our method surpasses the shortcomings of conventional methods such as DCP and CEP by incorporating a visibility improvement module, domain adaptation methods, and an optimized object detection model. By utilizing AOD-Net for image dehazing and GRL for domain adaptation, our model guarantees better

generalization over varying fog densities. Utilizing YOLOv8, boosted with CSP-Darknet53, SPPF, and CSP-PAN, provides high detection accuracy and strong feature extraction in low-visibility environments.

With thorough testing on a large and varied dataset, our model showed better performance in Mean Average Precision (mAP), F1 Score, Precision-Recall, and real-time inference speed (FPS) compared to current methods.

REFERENCES

- [1] Y. Song, T. Wang, P. Cai, S. K. Mondal, and J. P. Sahoo, "A comprehensive survey of few-shot learning: evolution, applications, challenges, and opportunities," *ACM Comput. Surv.*, 2023.
- [2] A. Moussaid *et al.*, "Citrus yield prediction using deep learning techniques: a combination of field and satellite data," *J. Open Innov. Technol. Mark. Complex.*, vol. 100075, 2023.
- [3] V. Voinarovska, M. Kabeshov, D. Dudenko, S. Genheden, and I. V. Tetko, "When yield prediction does not yield prediction: an overview of the current challenges," *J. Chem. Inf. Model.*, 2023.
- [4] K. V. N. Reddy and E. B. Reddy, "Crop yield prediction based on weather and soil parameters using regression tree model," in *Proc. 4th Int. Conf. Commun., Devices Comput. (ICCDC 2023)*, Springer Nature, 2023, p. 1.
- [5] U. Kaur, M. Kaur, and A. Khang, "Futuristic technologies in agriculture challenges and future prospects," in *Handb. Res. AI-Equipped IoT Appl. High-Tech Agric.*, pp. 347–364, 2023.
- [6] B. D. S. Barbosa, L. Costa, Y. Ampatzidis, V. Vijayakumar, and L. M. dos Santos, "UAV-based coffee yield prediction utilizing feature selection and deep learning," *Smart Agric. Technol.*, vol. 1, article 100010, 2021.
- [7] A. Moussaid *et al.*, "Machine learning applied to tree crop yield prediction using field data and satellite imagery: a case study in a citrus orchard," *Informatics*, MDPI, p. 80, 2022.
- [8] W. Zhang *et al.*, "Deep-learning-based in-field citrus fruit detection and tracking," *Hortic. Res.*, vol. 9, 2022.
- [9] K. Alibabaei, P. D. Gaspar, and T. M. Lima, "Crop yield estimation using deep learning based on climate big data and irrigation scheduling," *Energies*, vol. 14, no. 11, p. 3004, 2021.
- [10] Y. Liu, P. Sun, N. Wergeles, and Y. Shang, "A survey and performance evaluation of deep learning methods for small object detection," *Expert Syst. Appl.*, vol. 172, article 114602, 2021.
- [11] A. Moussaid, S. El Fkihi, and Y. Zennayi, "Citrus orchards monitoring based on remote sensing and artificial intelligence techniques: a review of the literature," in *Proc. 2nd Int. Conf. Adv. Technol. Humanity (ICATH)*, Rabat, Morocco, pp. 20–21, 2020.
- [12] P. Nevavuori, N. Narra, P. Linna, and T. Lipping, "Crop yield prediction using multitemporal UAV data and spatio-temporal deep learning models," *Remote Sens.*, vol. 12, no. 23, p. 4000, 2020.
- [13] W. Kurdthongmee, "A comparative study of the effectiveness of using popular DNN object detection algorithms for pith detection in cross-sectional images of parawood," *Heliyon*, vol. 6, no. 2, 2020.
- [14] M. I. Jordan, "Artificial intelligence—The revolution hasn't happened yet," *Harv. Data Sci. Rev.*, vol. 1, no. 1, pp. 1–9, 2019.